Vision-Language Models as Zero-Shot Pairwise Progress Rewards for Visual Robotic Reinforcement Learning

Leonardo Solidoro
leonardo.solidoro@tum.de

Tien Dung Pham
tien.dung.pham@tum.de

1. Introduction and Motivation

Reinforcement Learning (RL) requires a reward signal that tells the agent which behaviors should be reinforced. In many settings, this reward is either manually specified or learned from human feedback, both of which can be costly and difficult to obtain [3]. This problem is especially relevant in robotic manipulation, where different tasks require different definitions of progress. For example, a drawer-opening task may require measuring the drawer displacement, while an object-pushing task may require measuring the distance between the object and a target position. Although such quantities can often be accessed in simulation, they still require task-specific reward engineering. This makes reward design labor-intensive and difficult to transfer to settings where rewards should be inferred from visual observations, such as rendered frames or camera images.

Recent work has shown that pretrained vision-language models (VLMs) can provide reward signals for RL without task-specific reward training, by scoring how well an image observation matches a natural-language goal description [3]. Rocamonde et al. implement this idea with CLIP, computing the reward as the cosine similarity between the image embedding of the current observation and the text embedding of the goal description [3].

However, in robotic manipulation, a useful reward signal should capture not only how close the current state appears to the goal, but also whether meaningful progress has been made relative to an earlier state. For example, in drawer opening, the reward should reflect whether the drawer is more open than before, not just whether the current frame resembles the final goal. Similar ambiguity appears in multi-stage tasks such as pick-and-place, where grasping and lifting the object is meaningful progress but still less complete than placing it at the target. A single-frame score may therefore miss the relative ordering between intermediate stages.

This project addresses this limitation by using pairwise progress judgments instead of single-frame goal similarity. Given a before-and-after frame pair and a language goal, the VLM is asked to rate whether the later frame shows progress toward the task relative to the earlier frame. The model outputs a discrete progress score, which is used as a numerical reward signal for training an RL agent. We focus on large-scale frozen VLMs, mainly Qwen3-VL, and compare their pairwise progress scores against a CLIP-based single-frame reward baseline.

2. Research Question

The main research question is: Can a frozen VLM estimate manipulation task progress from before-and-after image pairs accurately enough to provide a useful reward signal for RL?

More specifically, this project asks:

1. How well do pairwise VLM progress scores correlate with ground-truth task rewards and benchmark success metrics?
2. Can pairwise VLM rewards rank successful rollouts above failed or partially successful rollouts?
3. Can an RL agent trained with pairwise VLM rewards achieve competitive task success compared to an agent trained with ground-truth rewards?
4. How do pairwise VLM rewards compare to single-frame CLIP-based rewards for robotic manipulation tasks?
5. In which cases do VLM progress judgments fail, such as visually similar states, subtle spatial differences, or multi-stage tasks?

3. Proposed Methodology

We propose a pairwise VLM progress reward. Given two rendered frames from a rollout, am earlier frame I_{t-k} and a later frame I_t, together with a language goal g, the VLM is prompted to rate how much progress the later frame shows toward the goal relative to the earlier frame. The proposed reward is:

$$r_t^{VLM} = \text{VLMProgress}(I_{t-k}, I_t, g) \tag{1}$$

The model outputs a discrete progress score:

$$r_t^{VLM} \in 0, 1, 2, \ldots, 10 \tag{2}$$

where 0 indicates no progress or negative progress, and 10 indicates clear progress toward task completion. This discrete reward format is motivated by RoboReward, which uses progress labels and reports that progress-based rewards are more useful for RL training than binary success rewards [2].

In practice, the VLM will be kept frozen and queried with a fixed prompt containing the goal description, the earlier frame, the later frame, and the scoring rubric. We will evaluate Qwen3-VL as the main pairwise VLM reward model and compare it against a CLIP-based single-frame reward baseline.

4. Experimental Setup

The main benchmark for this project will be Meta-World, a simulated robotic manipulation benchmark with 50 tasks using a shared Sawyer robot setup [5]. We focus on Meta-World because it provides diverse manipulation tasks, ground-truth task rewards, and binary success metrics, making it suitable for evaluating VLM-based reward signals.

The initial Meta-World task set will include tasks of increasing difficulty:

- Basic manipulation: reaching and pushing.
- Object interaction: drawer opening, door opening, and button pressing.
- More complex manipulation: peg insertion and pick-and-place.

If time permits, ManiSkill3 will be used as an additional generalization benchmark [4]. Candidate ManiSkill3 tasks include lift cube, turn faucet, open cabinet door, open cabinet drawer, and push chair.

For each task, we will collect rollouts with different levels of task completion. These will include expert rollouts, near-expert rollouts generated by adding action noise or stopping episodes early, and random rollouts that provide failure cases. This dataset will be used for offline reward evaluation before full RL training.

4.1. Reinforcement Learning Training Plan

The second stage of the project will train an RL agent using Soft Actor-Critic (SAC), replacing the environment reward with the proposed pairwise VLM progress reward [1]. At each reward-query step, the VLM compares a recent frame pair (I_{t-k}, I_t) together with the task goal g, and outputs a discrete progress score r_t^{VLM}.

Because VLM inference is expensive, rewards will not be computed at every environment step. Instead, we will compute VLM rewards periodically every k steps, for example $k = 5$. The reward will be assigned to the corresponding transition or held constant within the interval. This reduces inference cost while still providing a denser signal than only evaluating full episodes.

We will evaluate two frame-pair selection strategies. The first is a fixed-window strategy, where the VLM always compares the current frame I_t with the frame I_{t-k}. his provides a simple and consistent measure of local progress. The second is an adaptive reference-frame strategy for longer or multi-stage tasks. Instead of always using a fixed temporal offset, the reference frame can be updated when the rollout reaches a meaningful intermediate stage. For example, in a drawer-opening task, the comparison can first focus on reaching or grasping the handle, and later on pulling the drawer open. This may make the reward signal more informative when task progress consists of several sequential stages. We will compare the adaptive strategy against the fixed-window baseline to evaluate whether it improves reward quality and downstream policy learning.

5. Evaluation

We will compare three reward signals:

1. **Ground-truth environment reward:** The task-specific reward function provided by the simulation environment. This serves as the reference upper-bound reward signal.
2. **Single-frame CLIP reward:** A baseline reward computed from the similarity between the current image observation and the language goal, following prior work of Rocamonde et al. [3].
3. **Pairwise VLM progress reward:** The proposed method, where a VLM scores progress between an earlier and a later frame conditioned on the language goal.

5.1. Offline Reward Evaluation

Before full RL training, we will evaluate whether the proposed VLM reward provides a meaningful progress signal on pre-collected rollouts. The offline metrics will include:

- **Correlation with ground-truth rewards:** We will measure how well VLM progress scores correlate with environment rewards or task-success signals.
- **Rollout ranking accuracy:** We will test whether the reward assigns higher scores to successful or near-successful rollouts than to failed rollouts.
- **Stage ordering accuracy:** For rollouts with intermediate progress, we will test whether later, more complete stages receive higher scores than earlier stages.

5.2. RL Policy Evaluation

After training, policies will be evaluated using the benchmark's task success metrics. They include:

- **Success rate:** The fraction of episodes in which the agent achieves the task goal within the environment time limit. In Meta-World, success is usually defined by task-specific binary criteria, such as whether the relevant object reaches a target region within a fixed tolerance [5].

- **Sample efficiency:** Task success or average return as a function of environment interaction steps.
- **Comparison to baselines:** We will compare SAC trained with pairwise VLM rewards against SAC trained with ground-truth rewards and single-frame CLIP rewards.

6. Expected Contribution

This project will study whether pairwise VLM progress judgments can provide useful reward signals for visual robotic RL. The expected contributions are:

1. A reward-scoring pipeline that uses frozen VLMs to assign progress scores to before-and-after frame pairs conditioned on a language goal.
2. An empirical comparison between ground-truth environment rewards, single-frame CLIP-style rewards, and pairwise VLM progress rewards.
3. RL training experiments that evaluate whether SAC agents can learn from pairwise VLM rewards and achieve task success in simulation.
4. An analysis of failure cases, including visually similar states, subtle spatial differences, and multi-stage manipulation tasks.

References

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